Course syllabus

Jump to today

Miami University

College of Engineering & Computing

DEPARTMENT OF COMPUTER SCIENCE & SOFTWARE ENGINEERING

CSE 381: Systems 2: OS, Concurrency, Virtualization, and Security

ABOUT YOUR INSTRUCTOR

Alan Ferrenberg, Ph.D

- Email: alan.ferrenberg@miamioh.edu
- Office: 381 McVey Data Science Building
- Office hours:
  - Monday, Wednesday: 8:30 - 9:30 AM
  - Tuesday, Thursday: 12:00 - 1:00 PM

In a previous career, I was the Associate Vice President for Solution Delivery and Deputy CIO in IT Services, teaching in the evenings for the Department of Computer Science and Software Engineering (CSE). Since the Fall 2018 semester I've been teaching full-time for the CSE department where I'm an Associate Teaching Professor. I'm also an adjunct professor in the Department of Chemical, Paper and Biomedical Engineering. My background is in Simulational Physics and I was an active researcher from 1986 - 1999 until I moved to the dark side (management). I now do research as a hobby. My field of research is computer simulation and data analysis techniques, including the use of random numbers in simulations.

YOUR TAs

Section A: Max Schneider, Loc Tran

Section B: AJ Justiniano, Loc Tran

Section C: Max Schneider

COURSE DESCRIPTION

Overview

Introduction to operating systems concepts. The operating system as a resource manager. The principles for the design and implementation of operating systems. Process scheduling and deadlock prevention. Memory management, virtual memory, paging, and segmentation. Interrupt processing. Device management, I/O systems and I/O processing. Concurrency and multithreading. Virtualization and cloud services. Security and protection.

Class Sessions:

- Section A: Monday/Wednesday, 10:05 AM - 11:55 AM, Benton 006
- Section B: Monday/Wednesday, 12:10 PM - 2:00 PM, Benton 006
- Section C: Tuesday/Thursday, 2:15 PM - 4:05 PM, ENG 270

Prerequisites

CSE 278: Systems 1

Class Format

Class sessions will be a combination of brief lectures, demonstrations and in-class activities, some of which may be full-session labs or time to work on homework or group projects. The course grade will be determined through labs, homework assignments and exams.

Student Learning Objectives

By the end of this course, you should be able to:

1. Understand the key steps involved in loading and starting an operating system
2. Critique the design and tradeoffs in file systems and computer memory hierarchy and their interfaces
3. Use Multiprocessing and Inter-Process Communication (IPC) concepts and tools including pipes to monitor and control processes
4. Use Concurrency and Multithreaded programming concepts to create processes with multiple cooperating threads in ways that avoid deadlock and race conditions
5. Use virtualization and cloud services in effective ways
6. Use various security features provided by an operating system to avoid common security problems such as buffer overflow, stack smashing, trojans, and root kit problems

Key dates for the semester

Last day to Add/Drop through BannerWeb	Wednesday, January 31
Last day for Departments to add students to a course	Friday, February 2
Last day to drop without a grade	Thursday, February 15
Midterm exams	• February 19 or 20 • March 13 or 14 • April 22 or 23
Spring Break	March 25 - 31
Last day to drop with grade of W assigned	Monday, April 8
Final Exams (tentative)	• Section A: Wednesday, May 15 @ 10:15 AM - 12:15 PM • Section B: Wednesday, May 15 @ 12:45 PM - 2:45 PM • Section C: Tuesday, May 15 @ 12:45 PM - 2:45 PM

Course Topics

- Introduction and C++ review
- Intro to OS
- Processes, Fork and Exec
- Inter-Process Communication (IPC)
- Introduction to multithreading and synchronization-free threading
- Synchronization
- Filesystems
- Virtualization & Cloud Computing
- Cybersecurity

REQUIRED MATERIALS

Textbook

There is no single textbook for this course. There will be some reading assignments from textbooks available for free via Miami's subscription to **O'Reilly Books Online (https://learning.oreilly.com/home/)** but other prep material will come from websites and the in-class Google Slides presentations which will be made available in advance of class sessions.

Please see the **Course Resources (https://miamioh.instructure.com/courses/208923/pages/course-resources)** page for links to textbooks and other resources that could be useful.

ASSIGNMENT DESCRIPTIONS

Prep Work Quizzes

For most sessions there will be some kind of prep work (reading, videos, etc.) I expect you to do before class. When you have prep work, you'll also have a prep work quiz based on that material. You'll have two chances to take each prep work quiz and the higher grade will count. Feel free to work on these quizzes as you review the prep material.

Homework

There will be several homework assignments to be done outside of class time, though in some cases you may be able to begin the homework as part of an in-class activity. General homework policies:

- Homework will be assigned with a week or more of advance notice so you have time while the assignment is open to fit working on it into your schedule. **I will not accept late homework submissions.**
- All programming problems must be submitted via the Canvas CODE plug-in and must meet these requirements unless otherwise specified:
  - Must compile with no error messages and no warning messages
  - Must pass CSE style checking via the `cpplint.py` style checker
  - There may be other tests your program must pass in CODE before you will be able to submit it. **You should focus your initial attention on making these features work!**
- **Make sure you read that last bullet point: There are some simple requirements that programs must meet in order to be submitted. You should be checking your programs for these requirements continuously and ensuring that you fix any issues with them immediately.**
- Starting early is critical for successfully completing the homework. Canvas allows multiple submissions for a homework assignment and you should take advantage of that. Getting partial credit for an incomplete assignment is better than a zero!
- Possible exceptions to the "no late" homework policy are:
  - You have documented medical reasons clearly explaining the reason for missing deadlines. In the case of medical issues, you must notify me **well before the due date**, and I may ask for a doctor's note.
  - You have documented evidence indicating death or funeral in your immediate family (spouse, parents and grandparents, children and grand children, brothers and sisters, mother in law and father in law, brothers in law and sisters in law, daughters in law and sons in law).
- Your submission must be **your own work** unless the assignment is explicitly a team assignment. Submitting code from ANY other source is a violation of Miami's Academic Integrity policy. Some notes and clarifications:
  - It is OK to discuss issues with other students and to help each other through them, but **without** sharing code.
  - Similarly, you may **use online resources** to help with specific issues you are having, but submitting solutions you find online is a violation of Miami's Academic Integrity policy.
  - You may **consult** ChatGPT or other generative AI tools to help you with specific issues, but again, **submitting code that they generate and claiming it as your own is a violation of the Academic Integrity policy.**
- The documents describing lab and homework assignments give all of the requirements for your submission. Read them carefully and do what they says. **Attention to detail is very important**.
- Assignments may be graded on a randomly-selected subset of the requirements. So, for example, a shell script may be graded in some assignments and not in others. You will not know in advance which subset of requirements will be checked for an assignment so it is important to follow all of them. In some cases, you will know how many points various requirements are worth.
- Homework help via email: I will provide help and direction for programming projects via email. However, the following policy applies for obtaining help via email for homework related issues:
  - I will stop providing help via email 24 hours before the final deadline for a homework assignment. Bottom line: Start early and don't push deadlines!
  - Response to emails received after business hours (8-5), during holidays, and weekends will be sporadic. In particular, I am an early riser so I'm not likely to see emails sent in the evening or overnight until the next morning.

In-class Activities and Labs

There will be in-class activities in many sessions. Some will be graded and others not, and some will be brief and others will take up a full class session (a "lab"). As with homework assignments, any code you submit for a lab must be your own original work and the same policies for help from other students or other resources apply to labs.

Lab assignments are typically due at the end of the class session you work on them. Late lab submissions will not be accepted.

Exams

There will be three exams in this class held during class time on these dates:

Sections A and B:

- February 19
- March 13
- April 15

Section C:

- February 20
- March 14
- April 16

The only valid excuses for missing an exam are:

- You have documented medical reasons clearly explaining the reason for missing deadlines. You must let me know about a medical issue before the exam.
- You have documented evidence indicating death or funeral in your immediate family (spouse, parents and grandparents, children and grandchildren, brothers and sisters, mother in law and father in law, brothers in law and sisters in law, daughters in law and sons in law).
- If you require a makeup exam, you must obtain written (via email) permission from me no later than 24 hours prior to the test.
- Unexcused, missed exams will be recorded with zero points

The exams will be closed book and will be administered via Canvas. You will be permitted to bring one 8.5" x 11" page (front and back) of notes to use. I will use Proctorio for exams, **but only the browser lockdown feature**: Nothing will be recorded, nor will you be required to display your university ID. Proctorio will be set up simply to limit you to a single browser window for the exam.

Final Exam Policy: The University policy for the final exam is that no exam may be administered early except with written permission from the Associate Dean of the division. You will receive information about the process and deadlines later in the semester. This permission is typically not given except under extraordinary circumstances. Desire to leave town early is not an extraordinary circumstance. If you need to move your final exam to later in the week due to a conflict or excessive exams (per university policy), bring this to my attention as I have the ability to move a final exam to later in exam week.

Weekly Review Quizzes

Nearly every week you will have a take-home quiz on the material covered during the previous week. These will be due by midnight on Sunday. They will be true/false or multiple choice and you'll have two attempts at taking each quiz with the higher score being retained. These will be open-book quizzes.

GRADING INFORMATION

Determination of Grade

Category	Percentage of Grade
Exams	40%
Labs and Homework assignments	40%
Prep Work Quizzes	10%
Weekly Review Quizzes	10%

Letter Grade Definition

Final Grade	Percentage Range
A+	97 to 100%
A	93 to < 97%
A-	90 to < 93%
B+	87 to < 90%
B	83 to < 87%
B-	80 to < 83%
C+	77 to < 80%
C	73 to < 77%
C-	70 to < 73%
D+	67 to < 70%
D	63 to < 67%
D-	60 to < 63%
F	< 60%

COURSE EVALULATION

To help make sure that this course is effective in helping you learn the subject there will be multiple opportunities for you to provide anonymous evaluations or suggestions for improving the course.

- Retrospectives: Approximately every two weeks at the start of the semester you'll be given the opportunity to provide anonymous feedback via the following three questions:
  - What is interfering with your learning? (Things we should STOP doing.)
  - What suggestions do you have to improve your learning? (Things we should START doing.)
  - What am I doing that is helping you learn? (Things I should CONTINUE doing.)
- At the end of the semester there will be the normal final course evaluations. If you submit evidence that you did a final course evaluation (not your actual answers, just a copy of the page that shows you did an evaluation) you'll get extra credit.

ACADEMIC INTEGRITY

The Department of Computer Science and Software Engineering is committed to maintaining strict standards of academic integrity. The department expects each student to understand and comply with the **University's Policy on Academic Integrity** (http://www.miamioh.edu/integrity/): and Student Handbook (**Undergraduate** (https://miamioh.edu/policy-library/students/undergraduate/index.html), **Graduate** (https://miamioh.edu/policy-library/students/graduate/index.html) student policies).

Students may direct questions regarding academic integrity expectations to their instructor or to the department chair. All work submitted must be **original** for that class. Submitting the same project for two different classes is grounds for charging a student with academic misconduct unless prior written permission is received from **both** instructors.

In the CSE department, the penalty for an academic integrity violation is:

- A zero on the assignment in question
- A reduction of one letter in the final course grade

ADA & STUDENTS WITH DISABILITIES

Miami University is committed to ensuring equal access to students with disabilities. Miami's Office of Student Disability Services (SDS) assists students with determining eligibility for services and accommodation planning. Students who are entitled to disability-related academic adjustments, auxiliary aids, etc., must **register with SDS** (https://miamioh.edu/student-life/sds/student-tools/how-to-register/index.html) to receive accommodations in university courses. Please understand that formal communication from SDS must be presented prior to the coordination of accommodations for this course. For more information, students may contact SDS at (513) 529-1541 or via email at **sds@miamioh.edu** (mailto:sds@miamioh.edu).

To view the accessibility statement from Canvas follow **this link** (https://training.instructure.com/courses/347469/pages/accessibility-statement%20).

MISCELLANEOUS

COVID-19

We will follow University masking requirements which may change during the semester. As of this writing, masks are not required. You can review the situation and recommendations/requirements at Miami via the **University COVID-19 website** (https://miamioh.edu/covid19/).

Student Responsibilities

According to the Miami University Student Handbook, student responsibilities include:

- Attending class in accordance with University regulations;
- Being attentive during class, and participating constructively in class discussions;
- Doing the work assigned in each course to the best of his or her ability, and submitting it on time;
- Showing respect for instructors and fellow students at all times;
- Abiding by University regulations prohibiting academic misconduct (see Chapter 5 of Part 1 Undergraduate Academic Regulations);
- Endeavoring to ensure that the learning environment is free from all forms of prejudice that negatively influence student learning, such as those based on age, ethnicity, gender, mental or physical impairment, race, religion, sexual orientation, or gender identity.

During class students must avoid disruptive behavior which can result in student being removed from class. Disruptive behavior is defined as any behaviors that hamper the ability of instructor to teach or students to learn. Common examples of disruptive behaviors include, but are not limited to:

- Talking when the instructor or others are speaking
- Overt inattentiveness (e.g., sleeping or reading the paper in class)
- Creating excessive noise
- Entering the class late or leaving early
- Interrupting instructor during lecture with irrelevant comments
- Use of cell phones in the classroom
- Use of computing equipment for any purpose other than those pertaining to classroom activities

Attendance

This is what Miami's policy on student attendance says:

"Every student is expected to attend every class session for which the student is duly registered. It is the prerogative of the individual faculty member to set attendance policy for each individual course, and it is the responsibility of the individual faculty member to inform students of that policy in the course syllabus or other written document within the first week of the semester."

Here a link to **Miami's full policy on attendance** (https://miamioh.edu/policy-library/students/undergraduate/academic-regulations/class-attendance.html). In this class, I am not taking attendance and it is not a factor in your grade. However, I encourage you to attend all class sessions as there will often be demos as well as graded in-class assignments and labs that you will be expected to submit before the end of a class session.

Office Hours

I have office hours on four different days each week to help you with problems relevant to this course:

- Monday, Wednesday: 8:30 - 9:30 AM
- Tuesday, Thursday: 12:00 - 1:00 PM

If you cannot visit me during my office hours, you are welcome to contact me and make an appointment for a more suitable time. If you have difficulties with homework problems, please do not expect me to give you the solutions; that would defeat the purpose of homework assignments. I will, however, provide you with hints.

E-mail Communication

E-mail is the best way to get in touch with me as I will often be out of my office. I always respond to emails and will try do so promptly. I am **not** a night owl, so e-mails sent in the evening likely won't get a response until the next morning. And remember that email support for homework assignments ends 24 hours before the due date.

Resources and Support for Students

As an instructor, I have a **duty to report** (https://www.miamioh.edu/policy-library/employees/general-employment/non-discrimination/duty-to-report.html) . This means I am required to promptly report to the Deputy Title IX Coordinator (**titleix@miamioh.edu** (mailto:titleix@miamioh.edu)) any information a student shares with me regarding harassment, discrimination, sexual misconduct and interpersonal violence, or retaliation. **A report does not initiate an investigation. It engages a discussion of your resources, supportive measures, and options available.** If students want to speak with someone confidentially, the following resources are available on and off campus:

- **Student Health Services** (https://gnldr.online/tracker/click?redirect=https%3A%2F%2Fmiamioh.edu%2Fstudent-life%2Fstudent-health-service%2Findex.html&dID=1578667353841&linkName=Student%20Health%20Services) , **(513) 529-3000** (https://www.google.com/search?q=Miami+University+Student+Health&rlz=1C1GCEB_enUS814US814&oq=Miami+University+Student+Health&aqs=chrome..69i57j46i39i175i199j0l3j69i60l2j69i61.56858#)
- **Student Counseling Services** (https://gnldr.online/tracker/click?redirect=https%3A%2F%2Fmiamioh.edu%2Fstudent-life%2Fstudent-counseling-service%2F&dID=1578667353841&linkName=Student%20Counseling%20Services) , **(513) 529-4634** (https://www.google.com/search?rlz=1C1GCEB_enUS814US814&sxsrf=ALeKk02sACQS4DV0liKgWpTJGcUqvNjG1Q%3A1603376956794&ei=PJeRX8z8L4iPtAaSgpioDQ&q=Miami+University+Studer ECaABwAngAgAFbiAH_CpIBAjE4mAEAoAEBqgEHZ3dzLXdpesgBCMABAQ&sclient=psy-ab&ved=0ahUKEwjMis3PtMjsAhWIB80KHRIBBtUQ4dUDCA0&uact=5#)
- Women Helping Women (WHW) Sexual and Interpersonal Violence Support Specialists are available to support all students and can be contacted by emailing mu@womenhelpingwomen.org. As well as calling/texting 513-846-8402 between 9AM-5PM. The 24-hour hotline is 513-381-5610. WHW supports ALL survivors of dating/domestic violence, sexual assault, and stalking, regardless of gender identity or sexual orientation.

Speaking with a confidential resource person does not preclude students from making a formal report to the University if and when they are ready.

https://miamioh.edu/diversity-inclusion/programs-resources/report-incident/index.html (https://miamioh.edu/diversity-inclusion/programs-resources/report-incident/index.html)

For more information, please visit **https://miamioh.edu/campus-safety/sexual-assault/** (https://miamioh.edu/campus-safety/sexual-assault/) and **https://www.miamioh.edu/diversity-inclusion/oeeo/index.html** (https://www.miamioh.edu/diversity-inclusion/oeeo/index.html) .

Miami University Learning Community

Miami University is committed to fostering a supportive learning environment for all students irrespective of individual differences in gender, race, national origin, religion, handicapping condition, sexual preference, or age. Students should expect, and help create a supportive learning environment free from all forms of prejudice. Disparaging comments, sexist or racist humor, or questioning the academic commitment of students based upon these individual differences undermines our learning community. If such behaviors occur in class, please seek the assistance of your instructor or department chair.

A Final Note

It is my goal and Miami's to see you succeed academically. If you feel like you could use some help, in this course or others or more generally, please don't hesitate to reach out to me. And here are some Miami resources you can take advantage of:

- **Miller Center for Student Disability Services** (https://miamioh.edu/student-life/sds/)
- **Student Counseling Services** (http://miamioh.edu/student-life/student-counseling-service/index.html)
- **Student Health Service** (http://miamioh.edu/student-life/student-health-service/index.html)
- **Rinella Learning Center** (http://miamioh.edu/student-life/rinella-learning-center/index.html) , including tutoring services
- **Howe Center for Writing Excellence** (http://miamioh.edu/hcwe/)
- **English Language Learner Writing Center** (http://miamioh.edu/ellwc/)

See the **Personal Support web page** (http://miamioh.edu/student-life/student-affairs/care-and-concern/personal-support/index.html) from the Division of Student Life for links to additional resources.

Course summary:

Date	Details	Due
Mon, 29 Jan 2024	Prep for Session 01 (https://miamioh.instructure.com/courses/208923/assignments/2630458)	due by 10:05
	Quiz on Session 01 Prep (https://miamioh.instructure.com/courses/208923/assignments/2635370)	due by 10:05
Wed, 31 Jan 2024	Prep for Session 02 (https://miamioh.instructure.com/courses/208923/assignments/2632429)	due by 10:05
	Quiz on Session 02 Prep (https://miamioh.instructure.com/courses/208923/assignments/2687617)	due by 10:05
	Lab 00: Install Fedora and Visual Studio Code (https://miamioh.instructure.com/courses/208923/assignments/2667336)	due by 11:55
Sun, 4 Feb 2024	Quiz on Week 01 Content (https://miamioh.instructure.com/courses/208923/assignments/2699453)	due by 23:59
Mon, 5 Feb 2024	Prep for Session 03 (https://miamioh.instructure.com/courses/208923/assignments/2696832)	due by 10:05
	Quiz on Session 03 Prep (https://miamioh.instructure.com/courses/208923/assignments/2715212)	due by 10:05
Wed, 7 Feb 2024	Prep for Session 04 (https://miamioh.instructure.com/courses/208923/assignments/2715920)	due by 10:05
	Quiz on Session 04 Prep (https://miamioh.instructure.com/courses/208923/assignments/2716756)	due by 10:05
	Lab 01: Iterative Programming, Part A (https://miamioh.instructure.com/courses/208923/assignments/2709520)	due by 11:55
	Lab 01: Iterative Programming, Part B (https://miamioh.instructure.com/courses/208923/assignments/2709591)	due by 11:55
	Lab 01: Iterative Programming, Part C (https://miamioh.instructure.com/courses/208923/assignments/2709610)	due by 11:55
Sun, 11 Feb 2024	Quiz on Week 02 Content (https://miamioh.instructure.com/courses/208923/assignments/2718502)	due by 23:59
Mon, 12 Feb 2024	Prep for Session 05 (https://miamioh.instructure.com/courses/208923/assignments/2719711)	due by 10:05
	Quiz on Session 05 Prep (https://miamioh.instructure.com/courses/208923/assignments/2719755)	due by 10:05
Wed, 14 Feb 2024	Prep for Session 06 (https://miamioh.instructure.com/courses/208923/assignments/2720846)	due by 10:05
	Quiz on Session 06 Prep (https://miamioh.instructure.com/courses/208923/assignments/2720851)	due by 22:05
Sun, 18 Feb 2024	Quiz on Week 03 Content (https://miamioh.instructure.com/courses/208923/assignments/2722083)	due by 23:59
Mon, 19 Feb 2024	Test Proctorio for Midterm Exam 1 (Secure Browser) (https://miamioh.instructure.com/courses/208923/assignments/2722374)	due by 10:05
	202420-CSE381 Midterm Exam 1 (Secure Browser) (https://miamioh.instructure.com/courses/208923/assignments/2722573)	due by 11:10

Date	Details	Due
Wed, 21 Feb 2024	Prep for Session 08 (https://miamioh.instructure.com/courses/208923/assignments/2722577)	due by 10:05
	Quiz on Session 08 Prep (https://miamioh.instructure.com/courses/208923/assignments/2723004)	due by 10:05
Sun, 25 Feb 2024	Quiz on Week 04 Content (https://miamioh.instructure.com/courses/208923/assignments/2723997)	due by 23:59
Mon, 26 Feb 2024	Linux Driver's License (https://miamioh.instructure.com/courses/208923/assignments/2723998)	due by 10:05
	Prep for Session 09 (https://miamioh.instructure.com/courses/208923/assignments/2724366)	due by 10:05
	Quiz on Session 09 Prep (https://miamioh.instructure.com/courses/208923/assignments/2725172)	due by 10:05
	Lab 02: Linux Practice (https://miamioh.instructure.com/courses/208923/assignments/2725854)	due by 11:55
Wed, 28 Feb 2024	Prep for Session 10 (https://miamioh.instructure.com/courses/208923/assignments/2725928)	due by 10:05
	Quiz on Session 10 Prep (https://miamioh.instructure.com/courses/208923/assignments/2726018)	due by 10:05
Fri, 1 Mar 2024	Homework 01: Implementing an Inventory class in C++ (https://miamioh.instructure.com/courses/208923/assignments/2723499)	due by 23:59
Sun, 3 Mar 2024	Quiz on Week 05 Content (https://miamioh.instructure.com/courses/208923/assignments/2727034)	due by 23:59
Mon, 4 Mar 2024	Prep for Session 11 (https://miamioh.instructure.com/courses/208923/assignments/2727036)	due by 10:05
	Quiz on Session 11 Prep (https://miamioh.instructure.com/courses/208923/assignments/2727314)	due by 10:05
Wed, 6 Mar 2024	Prep for Session 12 (https://miamioh.instructure.com/courses/208923/assignments/2727037)	due by 10:05
	Quiz on Session 12 Prep (https://miamioh.instructure.com/courses/208923/assignments/2727475)	due by 10:05
Sun, 10 Mar 2024	Quiz on Week 06 Content (https://miamioh.instructure.com/courses/208923/assignments/2728471)	due by 23:59
Mon, 11 Mar 2024	Lab 03: Programmatic Permission String (https://miamioh.instructure.com/courses/208923/assignments/2731500)	due by 11:55
Wed, 13 Mar 2024	Proctorio Setup for Midterm 2 (Secure Browser) (https://miamioh.instructure.com/courses/208923/assignments/2722375)	due by 10:05
	202420-CSE381 Midterm Exam 2 (Secure Browser) (https://miamioh.instructure.com/courses/208923/assignments/2732654)	due by 11:55
Mon, 18 Mar 2024	Prep for Session 15 (https://miamioh.instructure.com/courses/208923/assignments/2731043)	due by 10:05
	Quiz on Session 15 Prep (https://miamioh.instructure.com/courses/208923/assignments/2732699)	due by 10:05
Wed, 20 Mar 2024	Prep for Session 16 (https://miamioh.instructure.com/courses/208923/assignments/2734069)	due by 10:05

Date	Details	Due
	Quiz on Session 16 Prep (https://miamioh.instructure.com/courses/208923/assignments/2734122)	due by 10:05
Sun, 24 Mar 2024	Quiz on Week 08 Content (https://miamioh.instructure.com/courses/208923/assignments/2734138)	due by 23:59
Mon, 1 Apr 2024	Prep for Session 17 (https://miamioh.instructure.com/courses/208923/assignments/2736079)	due by 10:05
	Quiz on Session 17 Prep (https://miamioh.instructure.com/courses/208923/assignments/2737059)	due by 10:05
Wed, 3 Apr 2024	Prep for Session 18 (https://miamioh.instructure.com/courses/208923/assignments/2738580)	due by 10:05
	Quiz on Session 18 Prep (https://miamioh.instructure.com/courses/208923/assignments/2738582)	due by 10:05
Fri, 5 Apr 2024	Homework 02: Synchronization-free Multithreading (https://miamioh.instructure.com/courses/208923/assignments/2738284)	due by 23:59
Sun, 7 Apr 2024	Quiz on Week 09 Content (https://miamioh.instructure.com/courses/208923/assignments/2738721)	due by 23:59
Mon, 8 Apr 2024	Prep for Session 19 (https://miamioh.instructure.com/courses/208923/assignments/2741495)	due by 10:05
	Quiz on Session 19 Prep (https://miamioh.instructure.com/courses/208923/assignments/2741645)	due by 10:05
	Lab 04: Multithreaded Banking System (https://miamioh.instructure.com/courses/208923/assignments/2741836)	due by 23:59
Wed, 10 Apr 2024	Prep for Session 20 (https://miamioh.instructure.com/courses/208923/assignments/2741722)	due by 10:05
	Quiz on Session 20 Prep (https://miamioh.instructure.com/courses/208923/assignments/2749244)	due by 10:05
Sun, 14 Apr 2024	Quiz on Week 10 Content (https://miamioh.instructure.com/courses/208923/assignments/2749770)	due by 23:59
Mon, 15 Apr 2024	Prep for Session 21 (https://miamioh.instructure.com/courses/208923/assignments/2751086)	due by 10:05
	Quiz on Session 21 Prep (https://miamioh.instructure.com/courses/208923/assignments/2751094)	due by 10:05
Wed, 17 Apr 2024	Lab 05a - sleep-wake (https://miamioh.instructure.com/courses/208923/assignments/2752545)	due by 12:05
	Lab 05b - busy-wait (https://miamioh.instructure.com/courses/208923/assignments/2752546)	due by 12:05
Sun, 21 Apr 2024	Quiz on Week 11 Content (https://miamioh.instructure.com/courses/208923/assignments/2752170)	due by 23:59
Mon, 22 Apr 2024	202420-CSE381 Midterm Exam 3 (Secure Browser) (https://miamioh.instructure.com/courses/208923/assignments/2753884)	due by 11:25
Wed, 24 Apr 2024	Prep for Session 24 (https://miamioh.instructure.com/courses/208923/assignments/2753394)	due by 10:05
	Quiz on Session 24 Prep (https://miamioh.instructure.com/courses/208923/assignments/2754384)	due by 10:05

Date	Details	Due
	Lab 06: Ell Ess - something like ls (https://miamioh.instructure.com/courses/208923/assignments/2755067)	due by 11:55
Sun, 28 Apr 2024	Quiz on Week 12 Content (https://miamioh.instructure.com/courses/208923/assignments/2755307)	due by 23:59
Mon, 29 Apr 2024	Prep for Session 25 (https://miamioh.instructure.com/courses/208923/assignments/2755361)	due by 10:05
	Quiz on Session 25 Prep (https://miamioh.instructure.com/courses/208923/assignments/2757023)	due by 10:05
	Lab07: Links (https://miamioh.instructure.com/courses/208923/assignments/2757121)	due by 11:55
Wed, 1 May 2024	Prep for Session 26 (https://miamioh.instructure.com/courses/208923/assignments/2757286)	due by 10:05
	Quiz on Session 26 Prep (https://miamioh.instructure.com/courses/208923/assignments/2757287)	due by 10:05
Sun, 5 May 2024	Quiz on Week 13 Content (https://miamioh.instructure.com/courses/208923/assignments/2757729)	due by 23:59
Mon, 6 May 2024	Prep for Session 27 (https://miamioh.instructure.com/courses/208923/assignments/2759951)	due by 10:05
	Quiz on Session 27 Prep (https://miamioh.instructure.com/courses/208923/assignments/2759953)	due by 10:05
Wed, 8 May 2024	Homework 03: Something like top (https://miamioh.instructure.com/courses/208923/assignments/2757205)	due by 23:59
Sun, 12 May 2024	Quiz on Week 14 Content (https://miamioh.instructure.com/courses/208923/assignments/2760512)	due by 23:59
Wed, 15 May 2024	202420-CSE381 Final Exam (https://miamioh.instructure.com/courses/208923/assignments/2764994)	due by 12:15
	Kahoot Extra Credit - First Chunk (https://miamioh.instructure.com/courses/208923/assignments/2749567)	
	Kahoot Extra Credit - Second Chunk (https://miamioh.instructure.com/courses/208923/assignments/2749568)	
	Kahoot Extra Credit - Third Chunk (https://miamioh.instructure.com/courses/208923/assignments/2749569)	
	Test Proctorio for Midterm Exam 3 (Secure Browser) (https://miamioh.instructure.com/courses/208923/assignments/2753897)	